



APPENDIX B

Code Metadata



First learn computer science and all the theory. Next develop a programming style.

Then forget all that and just hack.

— *George Carrette* —



B.1 | Code Versions

Code Name	Version	Reference
HOKI	1.7.1	Stevance et al. (2020)
KAITIAKI	1.0.0	This work
MATPLOTLIB	3.8.2	Hunter (2007)
NUMBA	0.58.1	Lam et al. (2015)
NUMPY	1.26.3	Harris et al. (2020)
PANDAS	2.1.4	McKinney (2010)
PYTHON	3.11.5	-
SCIPY	1.11.4	Virtanen et al. (2020)
TAKAHE	1.0.0	Richards et al. (2023)
TUI	8	Ghodla et al. (2023)*

Table B.1: The code versions we use in this thesis. *TUI was described in earlier papers, but [Ghodla et al. \(2023\)](#) was the first to name it.

B.2 | Parameter Configuration

In this appendix we list the default parameters chosen for our binary evolution runs, as detailed in subsection 2.6.1.

The possible mass loss prescriptions (IML1/IML2) are:

0. No mass loss
1. [Reimers \(1975\)](#)
2. [Blöcker \(1995\)](#)
3. [Vassiliadis & Wood \(1993\)](#)
4. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with interpolation errors described in [Richards et al. \(2024\)](#)
5. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) (described in [Eldridge et al., 2017](#))
6. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with slowly turning on WR rates described in [Richards et al. \(2024\)](#)
7. [de Jager et al. \(1988\)](#); [Nugis & Lamers \(2000\)](#); [Vink et al. \(2001\)](#) with no WR rates, described in [Richards et al. \(2024\)](#)
8. Unimplemented

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
NM2	199	Meshpoints	ICL	1		IT1	40	
ION	2		IT2	20		IAM	1	
JIN	15		IOP	5		OUT	15	
INUC	0		NCH	3		IBC	0	
JP	1		ICN	0		ITH	1	
IML1	5		IX	1		IML2	5	Mass loss (*2)
IY	1		ISGTH	0		IZ	1	
IMO	0		IMODE	2		IDIFF	0	
NWRT1	100		EPS	10^{-6}		NWRT2	1	
DEL	0.01		NWRT3	2		DHO	10^{-8}	
NWRT4	1		DT3	0		NWRT5	0	
DDD	1.0		NSAVE	10		NMONT	0	
DT1	0.1		DT2	1.2		CT1	3.0	
CT2	3.0		CT3	0.05		CT4	0.01	
CT5	0.6		CT6	0.35		CT7	0.45	
CT8	10^{-4}		CT9	0.1		CT10	0.3	

Table B.2: Our general initial conditions. Unless otherwise described, these are assumed to be enabled for all runs.

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
ZS	0.02		ALPHA	1.925		CH	0.7	
CC	0.173		CN	0.053		CO	0.482	
CNE	0.099		CMG	0.038		CSI	0.08	
CFE	0.072		RCD	10^6		OS	0.12	
RML	0		RMG	0		ECA	0	
XF	0.1		DR	0.001		RMT	1000.0	
RHL	0		AC	0		AK1	1.0	
AK2	10^{-4}		ECT	0		TRB	0	
IRAM	1		IRS1	1		VROT1	0	
IRS2	1		VROT2	10.0		FMAC	1.0	
FAM	0		IVMC	0		TRC1	700.0	
IVMS	0		TRC2	0		MWTS	0	
IAGB	0		ISGFAC	0		FACSGMIN	0.01	
SGTHFAC	0		ISTART	0		HKH	1.0	
GFF	10.0	Removed	NNMOD	0		CEPR	0	CEE prescription
ALPHACE	1.0		-	-	-	-	-	-

Table B.2: Continued

9. ZAMS evolution to a specific ZAMS mass, custom routine by Jan Eldridge